Computers and Electrical Engineering 40 (2014) 59–69

Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compeleceng

Salt and pepper noise filtering with fuzzy-cellular automata ☆

U. Sahin [a,*], S. Uguz [b], F. Sahin [a]

[a] Multi Agent Biorobotic Laboratory, Rochester Institute of Technology, Rochester, NY 14623, USA
[b] Department of Mathematics, Harran University, Şanliurfa 63120, Turkey

A R T I C L E   I N F O

Article history:
Available online 9 December 2013

A B S T R A C T

A new image denoising algorithm is proposed to restore digital images corrupted by impulse noise. It is based on two dimensional cellular automata (CA) with the help of fuzzy logic theory. The algorithm describes a local fuzzy transition rule which gives a membership value to the corrupted pixel neighborhood and assigns next state value as a central pixel value. The proposed method removes the noise effectively even at noise level as high as 90%. Extensive simulations show that the proposed algorithm provides better performance than many of the existing filters in terms of noise suppression and detail preservation. Also, qualitative and quantitative measures of the image produce better results on different images compared with the other algorithms.

1. Introduction

Impulse noise in an image is present due to bit errors in transmission or introduced during the signal acquisition stage. This noise is caused by malfunctioning pixels in camera sensors, faulty memory locations in hardware, transmission in a noisy channel and external disturbance such as atmospheric disturbance [1]. A standard signal processing requirement is to eliminate randomly occurring impulses without disturbing edges and preserving the image integrity so that it can be later used for subsequent image processing operations such as edge detection, image segmentation and others.

There are two types of impulse noise, salt and pepper noise and random valued noise. For images corrupted by salt and pepper noise, the noisy pixels can take only the maximum and the minimum values in the dynamic range. Salt and pepper noise can significantly deteriorate the quality of an image. How to efficiently remove this kind of impulse noise is an important research task. There are many studies on the restoration of images corrupted by impulse noise. It is well known that linear filtering techniques are not effective in removing impulse noise when the noise is non-additive [2]. The median filter was once the most popular nonlinear filter for removing impulse noise because of its good denoising power and computational efficiency. However, the filter smears some of the details and the edges of the original image when the noise level is over 50 [3]. The switching approach algorithms have been introduced to protect the fine details in the image and to avoid the damage of the uncorrupted pixels [4]. Nonlinear filters such as adaptive median filter (AMF) [5] can be used for discriminating corrupted and uncorrupted pixels and then apply the filtering technique. Srinivasan and Ebenezer [2] presented an efficient decision-based algorithm (EDBA) in which the corrupted pixels are replaced by either the median pixel or neighborhood pixel by using a fixed window size of 3×3 resulting in lower processing time and good edge preservation. Although EDBA filter showed promising results, a smooth transition between the pixels is lost leading to degradation in the visual quality of the image in the form of line artifacts, since it only considers the left neighborhood from the last processed value. A fuzzy-based switching filter for restoring images corrupted with impulse and additive noise is introduced in [6] and in [7].

☆ Reviews processed and recommended for publication to Editor-in-Chief by Associate Editor Dr. Ziya Arnavut.

* Corresponding author. Tel.: +1 585 764 94 22.
E-mail addresses: useee@rit.edu (U. Sahin), selmanuguz@gmail.com (S. Uguz), feseee@rit.edu (F. Sahin).

0045-7906/$ - see front matter Published by Elsevier Ltd.
http://dx.doi.org/10.1016/j.compeleceng.2013.11.010

To address high noise density, a noise adaptive soft-switching median (NASM) filter was introduced in [8], which consists of a three-level hierarchical soft-switching noise detection process. Also, the adaptive switching median filter [9], or the median filter based on homogeneity information [10] has been proposed. In [11], highly effective impulse noise detection method (HEIND) is used for determining the corrupted pixels in the noisy image. Then an efficient direction based adaptive weighted switching median filter is applied on each corrupted pixels. These filters are good at detecting noise even at a high noise level. Most of them has main drawback due to the noisy pixels which is replaced by some median value in their vicinity without taking into account local features such as the possible presence of edges. Hence, details and edges are not recovered satisfactorily, especially when the noise level is high.

Cellular automata (CA) are discrete dynamic systems whose behavior is completely specified in terms of a local relation [12]. Space is represented by a uniform grid, with each site or cell containing a few bits of data. These data are also called a cells state. Time advances in discrete steps and the laws of the universe are expressed by a single recipe. At each step, every cell computes its new state from that of its close neighbors. Thus, the laws of the system are local and uniform. CA already supply many useful models for investigations in natural science, combinatorial mathematics and computer science, and as this paper will try to demonstrate, they can also be used as a very fast and efficient tool for image processing purposes [13–17]. For example, in [18], CA and fuzzy logic was used as a filtering tool for different type of impulse noise. However, the study is far from away to give sound mathematical background related with cellular automata. In this study, it is showed that CA is a very sound tool to use in image processing applications by giving exact mathematical representation. Fuzzy logic was also applied in [19]. They used an extra noise detection module before filtering which it was not needed in this study. Moreover, it was constructed a fuzzy rule base which puts some extra load for the performance. In our work, in terms of rule based systems, only one rule was used as a cellular automata transition rule.

In this paper, with the growing appeal of cellular automata, by combining with fuzzy theories, a new kind of impulse noise filter is improved. It exploits the fuzziness in the neighborhood of the corrupted pixel to produce local state cellular automaton transition rules. The algorithm used for noise elimination does not require any parameters to tune and thus no training stage is required. It can successfully cancel the noise from highly corrupted images very efficiently and preserves the edge details finely. Another advantage of the proposed fuzzy cellular automaton filter is that it uses a fixed window size of 3×3 for the noise filtering process. Also, this study presents the rigid mathematical background of CA for the image processing and filtering using windowing application that have not been given by any other CA studies applied to this field. The efficiency of the proposed algorithm is tested using standard images and compared with the other algorithms in a same way.

The rest of this paper is organized as follows. In Section 2 we provide a brief preliminaries of the cellular automata. In Section 3 we develop a new fuzzy cellular automata filter for noise elimination. Simulation results are presented in Section 4 by comparing the results with the other best-known algorithm. Finally, the paper is concluded in Section 5.

2. Technical preliminaries

2.1. Mathematical structure of cellular automata

In this section we present 2D CA with spin values in Z_{256} (i.e. spin values of gray scale image) by using a specific local nonlinear filtering rule. First, we recall the definition of a cellular automaton. We consider the 2-dimensional integer lattice Z^2 and the configuration space $\Omega = \{0,1,2,\ldots,255\}^{Z^2} = Z_{256}^{Z^2}$ with elements

$$\sigma : Z^2 \to \{0,1,2,\ldots,255\} = Z_{256}.$$

The value of σ at a point $n \in Z^2$ will be denoted by σ_n. Let $u_1,\ldots,u_s \in Z^2$ be a finite set of distinct vectors and

$$F : Z_{256}^s \to Z_{256}$$

a function. A CA with local rules F is defined as a pair (Ω, T), where the global transition map $T : \Omega \to \Omega$ is given by

$$(T\sigma)_n = F(\sigma_{n+u_1},\ldots,\sigma_{n+u_s}), \quad n \in Z^2.$$

The space Ω is assumed to be equipped with a (metrizable) Tychonoff topology. For a fixed $v \in Z^2$, let $U^v : \Omega \to \Omega$ be the shift map: $U^v(a) = b_m$, where $b_m = a_{m-v}$, for all $m \in Z^2$. A CA is a continuous map which commutes with all shifts. It is easily seen that the global transition map T introduced above and the shift operator U are continuous in this topology (see [20] for the details). The 2-D finite CA consists of $m \times n$ cells arranged in m rows and n columns, where each cell takes one of the values 0, 1, 2,…, or 255. A configuration of the system is an assignment of the states to all cells. Every configuration determines the next configuration via a linear transition rule that is local in the sense that the state of a cell at time $(t+1)$ depends only on the states of some of its neighbors at the time t using the local rules F. For 2D CA nearest neighbors, there are nine cells arranged in a 3×3 matrix centering that particular cell (see Table 1). For 2-D CA there are some classic types of neighborhoods, but in this work we only restrict ourselves to the adjacent neighbors. So, we can define the $(t+1)^{th}$ state of the (i,j)th cell as follows:

$$\begin{aligned} x_{(i,j)}^{(t+1)} &= F(x_{(i,j)}^{(t)}, \quad x_{(i+1,j)}^{(t)}, \quad x_{(i+1,j-1)}^{(t)}, \quad x_{(i,j-1)}^{(t)}, \quad x_{(i-1,j-1)}^{(t)}, x_{(i-1,j)}^{(t)}, \quad x_{(i-1,j+1)}^{(t)}, \quad x_{(i,j+1)}^{(t)}, \quad x_{(i+1,j+1)}^{(t)}), \\ &= F(x_{(i,j)}^{(t)}), \end{aligned} \tag{1}$$

Table 1
2D finite $CA_{3\times 3}$ with the center element $C_{i,j}$.

$C_{i-1,j-1}$	$C_{i-1,j}$	$C_{i-1,j+1}$
$C_{i,j-1}$	$C_{i,j}$	$C_{i,j+1}$
$C_{i+1,j-1}$	$C_{i+1,j}$	$C_{i+1,j+1}$

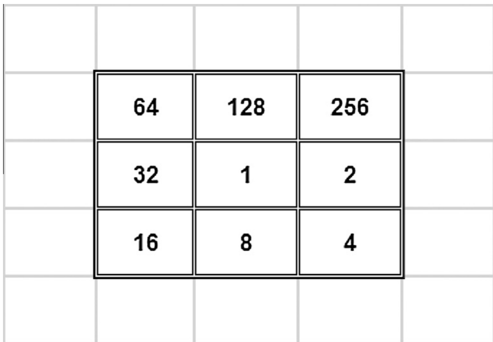

Fig. 1. 2D CA main rule box: from Rules 1 to 512 can be obtained by adding linearly of the fundamental rules: Rules 1, 2, 4, ..., 511.

where F can be defined in many different ways with respect to any model (see our F in the next section). The value of each cell for the next state may not depend upon all nine neighbors. The conventional method of defining a rule number for a linear rule in 2D CA can be explained in Fig. 1.

2.2. Interpreting the rules of CA

Using cellular automaton for image processing should involve the relation between the automaton and image processing. A digital image with size $m \times n$ is a bi-dimensional array which each element is called a pixel with a gray level or color value. Each pixel can be characterized by the triplet (i, j, k) where (i, j) represents its position in the array and k represents gray level or color. The image can be then considered as a particular configuration state of a cellular automaton that has as cellular space on the array defined by the image. Each site in the array corresponds to a pixel. cellular automaton growth is used for image processing and is controlled by predefined rules, usually controlled by program, or by some grammar rules.

CA consists of a regular grid of cells (pixels) and each cell (pixel) can be in only one of a finite number of possible states. For a gray level image, each cell (pixel) has 256 states that are composed of integer numbers from 0 to 255. The state of a cell (pixel) is determined by the previous states of a surrounding neighborhood of cells (pixels) and is updated synchronously in discrete time steps. The identical rules contained in each cell is essentially a finite state machine, usually specified in the form of grammar rules with an entry for every possible neighborhood configuration of states.

A specific rule convention that is adopted here is as follows. The central box (see Fig. 1) represents the current cell that is the cell under consideration and all other boxes represent the eight nearest neighbors of that cell. Each of the cells can be taken as a variable. Thus for 2D CA there are 9 variable to be considered. The number within each box represents the rule number characterizing the dependency of the current cell on that particular neighbor only. Rule 1 characterizes dependency of the central cell on itself alone where as such dependency only on its top neighbor is characterized by Rule 128, and dependency on all neighbors is characterized by Rule 510 (also called Moore neighborhood, see Fig. 2), and so on. These nine rules are called fundamental rules (see Fig. 1). In the case, the cell has dependency on two or more neighboring cells, the rule number will be the arithmetic sum of the numbers of the relevant cells. The number of rules generated by a combination of these 9-variables is $\binom{9}{0} + \binom{9}{1} + \ldots + \binom{9}{9} = 512$ which includes rule characterizing due to no dependency. If the next state of a cell depends on the current state of itself and its right neighbor, it is referred to as Rule 3 (=Rule 1 + Rule 2) since it is dependent only on two cells. If the next state depends on the present state of itself and all its Moore neighbors, it is referred to as Rule 511 (=Rule 1 + Rule 2 + Rule 4 + Rule 8 + Rule 16 + Rule 32 + Rule 64 + Rule 128 + Rule 256). All other linear rules can be obtained in the similar way.

3. Fuzzy cellular automata filter

Let $C_{i,j}$ represent the central pixel of Moore neighborhood in the selected cellular automata. Let N_{min} and N_{max} represent the values of the minimum and maximum intensity within a selected window around $C_{i,j}$. The following noise detection rules can be applied before filtering which is being used in conventional filter such as conservative smoothing filter [6],

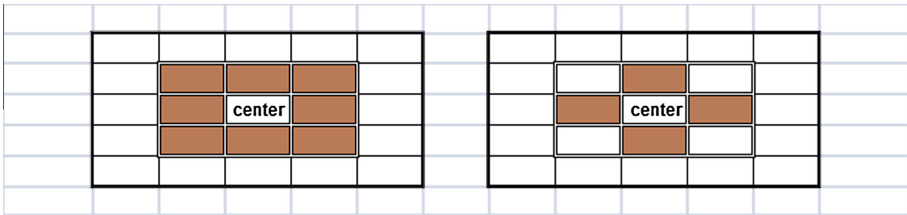

Fig. 2. The Moore (left) and von Neumann (right) neighborhoods of center cell. The nearest neighborhood comprises the center cell. These are the common CA neighborhoods which correspond to rule Rules 510 and 170 respectively.

Table 2
Different CA rules performance on images corrupted by 70% noise.

Rules	Bridge PSNR	Lena PSNR
Rule 45	22.71	25.28
Rule 79	22.73	26.21
Rule 290	23.21	27.86
Rule 312	24.04	29.08
Rule 2	15.93	15.62
Rule 165	23.43	27.19
Rule 170 (von Neumann)	23.96	28.02
Rule 510 (Moore)	24.14	29.27

Table 3
Comparison between the methods using Lena image, for different rate of impulse noise density for Rule 510 with 512 × 512 pixel images.

Noise ratio Method	10%	20%	30% PSNR	40% Value	50%	60%	70%
PSM	33.8	34.2	29.1	28.4	24.7	24.4	19.4
DDBSM	34.4	32.8	29.9	28.3	24.5	21.8	17.4
ISM	33.8	31.2	28.7	28.5	27.7	24.9	23.3
NASM	35.1	31.4	33.4	30.8	28.9	24.6	21.7
Srinivasan and Ebenezer	38.4	37.3	35.9	34.1	32.2	30.4	28.6
Ching et al.	41.4	38.2	35.9	34.	32.6	31.2	29.7
Chan et al.	42.6	39.3	37	34.3	31.8	30.8	29.7
Proposed	40.7	37.1	34.9	33.2	31.8	30.5	29.2

Table 4
SSIM values for Lena and Bridge images, for 70% impulse noise density and Rule 510 with 512 × 512 pixel images.

Method	Lena SSIM	Bridge SSIM
PSM	0.7	0.55
DDBSM	0.32	0.27
ISM	0.68	0.45
NASM	0.7	0.52
Srinivasan and Ebenezer	0.84	0.74
Ching et al.	0.74	–
Chan et al.	0.86	0.755
Proposed	0.88	0.757

if $N_{min} < C_{i,j} < N_{max}$, then $C_{i,j}$ is an uncorrupted pixel and its value is left unchanged. Otherwise $C_{i,j}$ is a noisy pixel.

After determination of the corrupted pixel, calculate the smallest and the largest values of the window around $C_{i,j}$ to restore the pixel including at least one of the N_{min} and N_{max} values by using minimum and maximum operator as follows:

$$\zeta_{i,j,k,l} = min\{C_{i+k,j+l} \neq 0\}$$

and

$$\xi_{i,j,k,l} = max\{C_{i+k,j+l} \neq 255\}$$

Table 5
Comparison between the methods using Bridge image, for different rate of impulse noise density for Rule 510 with 512 × 512 pixel images.

Noise ratio Method	10%	20%	30% PSNR	40% Value	50%	60%	70%
PSM	31.2	27.5	24.9	23.7	22.2	19.8	17
DDBSM	29.4	26.7	25.4	23.6	22.4	19.2	15.9
ISM	28.6	25.9	24.7	23.4	23.1	22.7	20.1
NASM	27.8	27.1	26.4	25.7	24.5	22.8	19.9
Srinivasan and Ebenezer	33.4	31.3	29.7	28.3	26.1	24.2	23.1
Chan et al.	37.1	33.8	32.5	30.1	28.1	26.7	25
Proposed	34.4	31.2	29.2	27.8	26.5	25.3	24.1

Table 6
Comparison between the methods using Lena image, for different rate of impulse noise density for Rule 510 with 256 × 256 pixel images.

Noise ratio Method	20%	30%	40% PSNR	50% Value	60%	70%	80%
Chan et al.	32.9	30.7	28.9	27.3	26.1	24.1	22.39
Srinivasan and Ebenezer	29.8	28.1	27.4	26.8	26.1	25.5	24.49
Wang et al.	34.3	32	30.2	28.5	27.3	26	24.53
Proposed	33.4	31.5	30	28.7	27.6	26.1	24.68

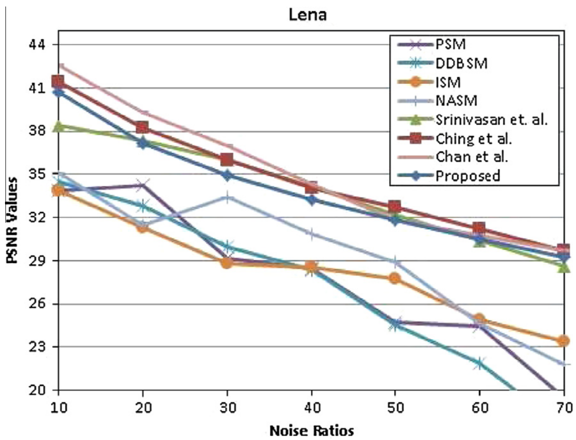

Fig. 3. PSNR values of restored Lena image for different algorithms.

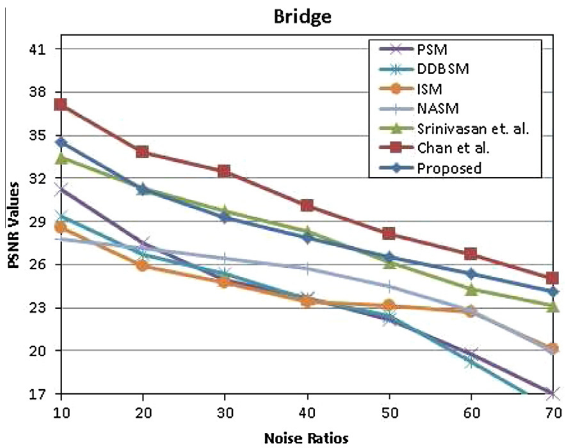

Fig. 4. PSNR values of restored Bridge image for different algorithms.

Fig. 5. Restored images of different filters for Lena image for 70% salt-and-pepper noise with 512 × 512 pixel image.

where $C_{i+k,j+l}$ represents the neighborhood pixels values in the same window and $k, l \in \{-1, 0, 1\}$. For the other values in the window, we can define a membership degree which measures their own contribution by using wisely selected gaussian membership function. It will assign a non-zero value for each pixel in that window and eventually all weighted average of the pixels around $C_{i,j}$ will be used for restoration.

By applying the gaussian membership function,

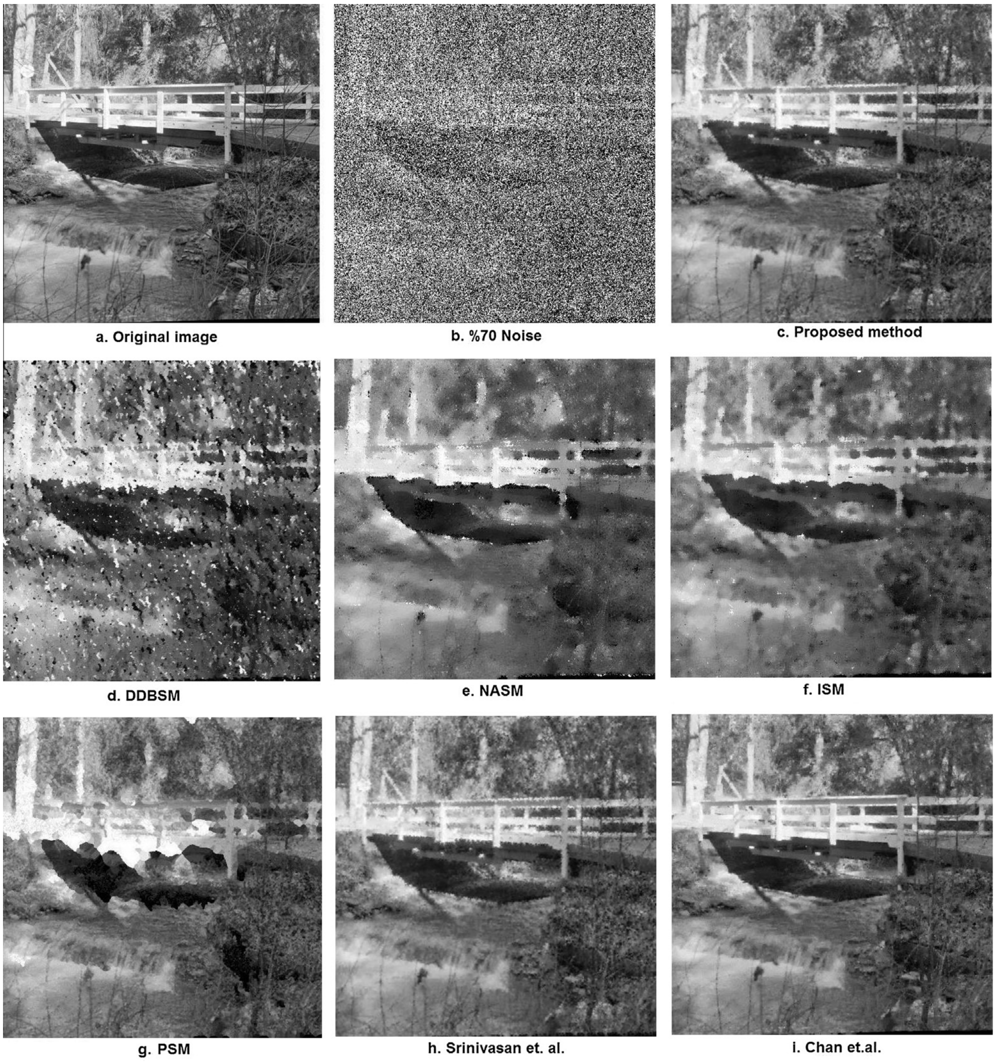

Fig. 6. Restored images of different filters for Bridge image for 70% salt-and-pepper noise with 512 × 512 pixel image.

$$\mu_{C_{i,j}}(C_{i+k,j+l}) = e^{-\frac{(C_{i+k,j+l}-\omega_{i,j,k,l})^2}{2\sigma^2_{i,j,k,l}}} \tag{2}$$

where $\omega_{i,j,k,l}$ and $\sigma_{i,j,k,l}$ are the mean and the standard deviation of the pixels in Moore neighborhood of the $C_{i+k,j+l}$, respectively, we can calculate how much contribution will be added by the neighborhood pixels until some degree. Finally, all these operations can be combined in a partial function to construct our local state rule, F of cellular automaton as follows:

$$F(C_{i,j}) = \begin{cases} \psi_{C_{i,j,k,l}} & , \exists\ C_{i,j} = 0\ or\ C_{i,j} = 255 \\ 127 & , \forall\ C_{i,j} = 0\ or\ C_{i,j} = 255 \\ \phi_{C_{i,j,k,l}} & , 0 < C_{i,j} < 255 \end{cases} \tag{3}$$

where

$$\psi_{C_{i,j,k,l}} = \frac{\zeta_{i,j,k,l} + \xi_{i,j,k,l}}{2}$$

and

$$\phi_{C_{i,j,k,l}} = \frac{\Sigma\mu_{C_{i,j}}(C_{i+k,j+l}) \cdot C_{i+k,j+l}}{\Sigma\mu_{C_{i,j}}(C_{i+k,j+l})}.$$

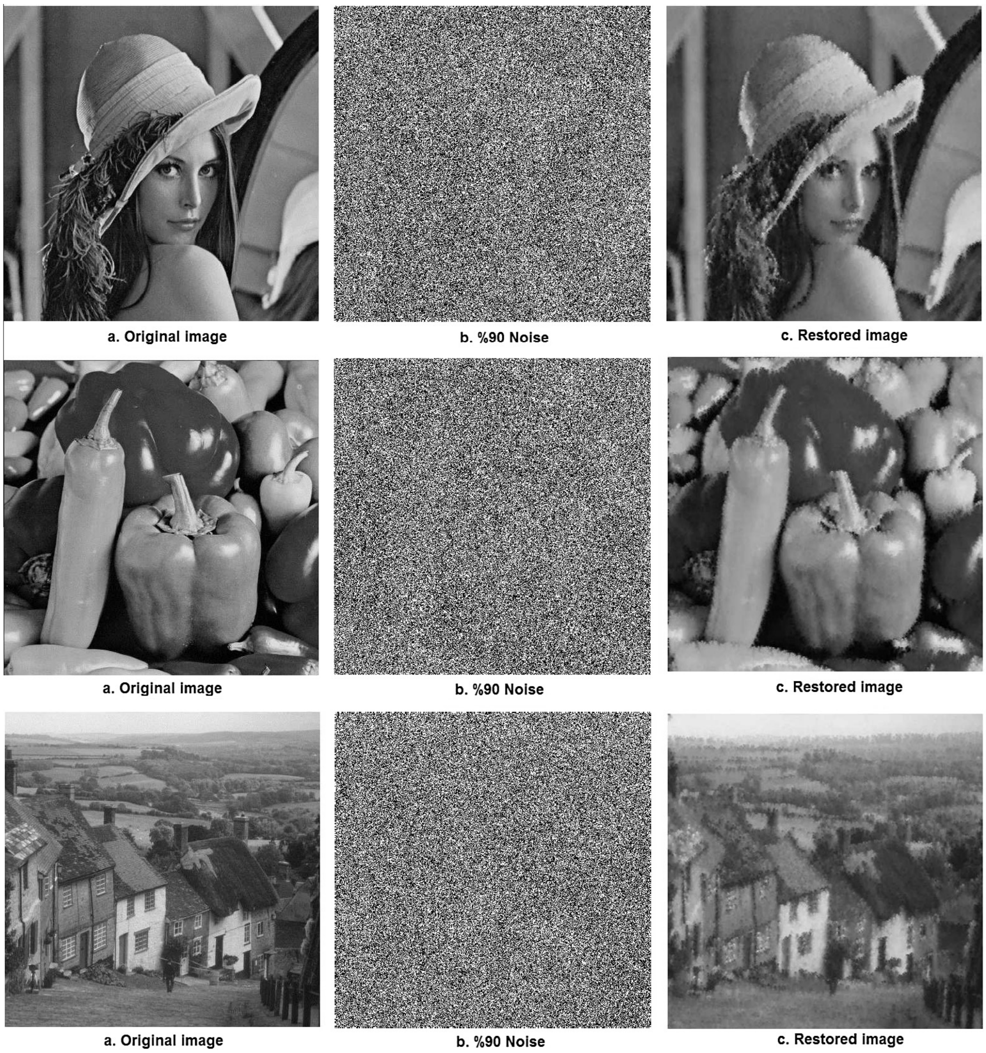

Fig. 7. Restored images of Lena, Pepper, Golden Hill images for 90% salt-and-pepper noise with 512 × 512 pixel image.

4. Experimental study

We have tested our method on the well known Lena and Bridge images, of size 512 × 512 and 256 × 256 pixels, contaminated by different magnitudes of salt and pepper noise. The noise levels are varied from 10% to 90% with increments of 10%, and restoration performances are quantitatively measured by the peak signal to noise ratio (PSNR) and Structural Similarity (SSIM) index calculated using the following standard formulas

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{255^2}{\mathrm{MSE}}\right) \tag{4}$$

$$\mathrm{MSE} = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij} - y_{ij})^2 \tag{5}$$

where mean square error (MSE) is the mean squared error between the original and filtered image, $m \times n$ is size of image, x_{ij}, y_{ij} denote the original and processed image pixels, respectively.

$$\mathrm{SSIM}(O,R) = \frac{(2\mu_O\mu_R + c_1)(2\sigma_{OR} + c_2)}{(\mu_O^2 + \mu_R^2 + c_1)(\sigma_O^2 + \sigma_R^2 + c_2)} \tag{6}$$

where μ_O and μ_R are the averages, σ_O and σ_R are the variances of original image (O) and restored images (R), respectively. σ_{OR} is the covariance of original and restored image and $c_1 = (k_1L)^2, c_2 = (k_2L)^2$ are two variables to stabilize the division with weak denominator, L is the dynamic range of the pixel-values and $k_1 = 0.01$ and $k_2 = 0.03$ by default [21]. From Table 2, it can be observed how cellular automata presents flexibility about the filtering approach by presenting different type of rules. One can easily obtain the rules and decides how the filtering will be performed based on PSNR, visual quality of the image and computation time of the CA rules. For instance, Rules 312 and 510 give better PSNR results over the other compared rules in Table 2. However, one can choose Rule 312 to decrease the computation time instead of Rule 510 without losing so much performance quality and visual image quality. The PSNR are calculated for proposed filter and a comparison of performance with the other filters, namely, NASM [8], ISM [23], DDBSM [22], PSM [24], Srinivasan and Ebenezer [2], Lu and Chou [25], Chan et al. [27], Wang and Wu [26] are presented in Tables 3–6.

Tables 3–5 present the proposed algorithm performance in terms of PSNR values and SSIM index. Figs. 3 and 4 show that the proposed filter provides better PSNR graphically when compared to the other algorithms except [25,27]. Figs. 5 and 6 also show the subjective visual qualities for qualitative analysis of the filtered images using various algorithms for the image “Lena” and “Bridge” images corrupted by 70% impulse noise. From Table 4, it can also be seen that the proposed filter gives better SSIM index values for the Lena images compared to the other algorithms. However, for images of size 256 × 256 pixels, it can be seen from Table 6 that the proposed method outperforms all the other methods. Larger images contain much more information around the same pixel when compared to one of the same small images and the windows around the pixels that consist of different pixel values. When the filtering algorithms are applied to the same image of different size, it is normal to obtain varied values. With respect to our algorithm, the fuzzy cellular transition rule tends to compute the weighted average of the pixel values of the windows. Also, in a smaller image, center values of the windows will be closer to the surrounded values than the larger ones. Hence, small size of the same image illustrates better results.

From Fig. 7, at the high noise probabilities (90%) the visual qualities of the restored images are very good considering the abundance of image details. It removes almost all of the noise pixels while preserving image details very well. For image processed with DDBSM, NASM and ISM from the subjective quality of the image one can observe that some of the impulse noises are not fully suppressed while blurring and distortion are serious. PSM and NASM suppress much of the noise where they fails to preserve edge information while maintaining details of the image.

Fig. 8 also shows the subjective visual qualities of the filtered images using various algorithms for the Lena image corrupted by 80% impulse noise with 256 × 256. Fig. 9 shows that the proposed filter provides better PSNR graphically when compared to the other well-known and most cited algorithms. The proposed algorithm in this paper outperform other algorithms in respect to PSNR and the subjective visual qualities especially for relatively small images. In particular, as noise density becomes high and images are greatly corrupted by the impulse noise, fuzzy cellular automata filter shows remarkable results. The fuzzy cellular automaton filter also uses a fixed 3 × 3 window for processing, leading to much smaller computation time.

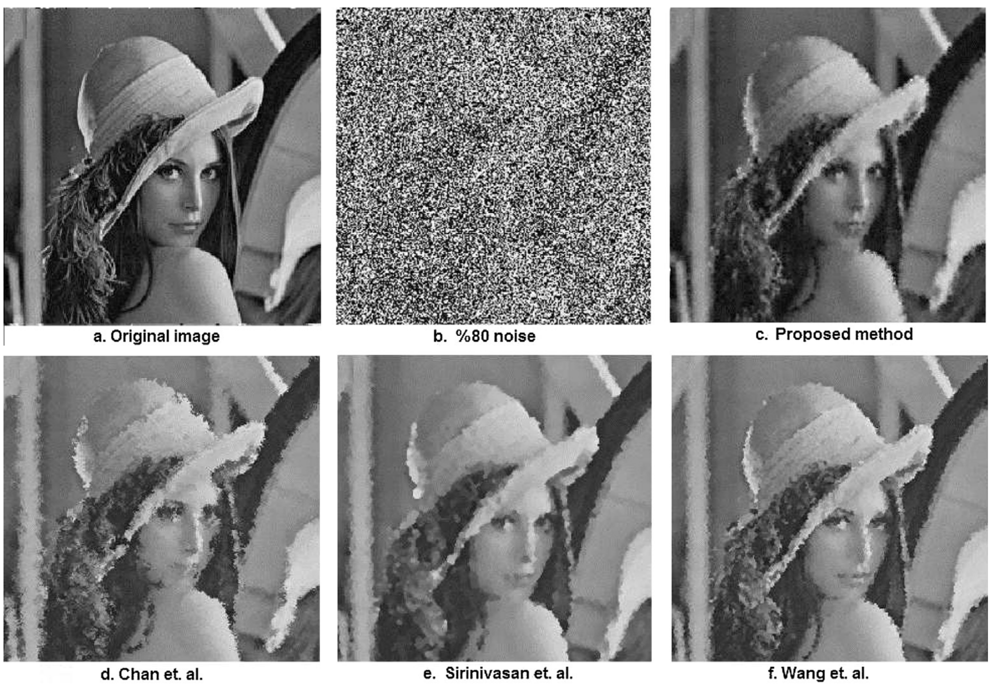

Fig. 8. Restored Lena image for different algorithms with 256 × 256 pixel images for 80% salt-and-pepper noise.

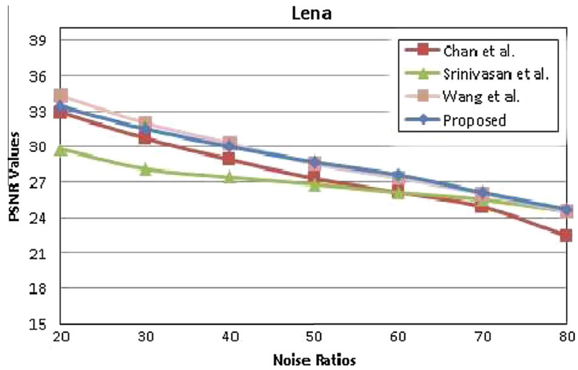

Fig. 9. PSNR values of restored Lena image for different algorithms with 256 × 256 pixel image.

5. Conclusion

Cellular automata is a new theoretical tool for implementing image processing tasks. In this paper, the cellular automata for image denoising is discussed, and a new local transition function based on fuzzy theory is proposed, which can filter image salt and pepper noises effectively, especially when noise densities is ranged from different scales. The proposed filter shows consistent and stable performance across a wide range of noise densities varying 10%-90%. Effective noise removal can be observed up to 90% while preserving edges finely. Several modifications to the standard cellular automata formulation can be applied to improve the filtering performance for different type of noises in the next studies.

Acknowledgment

S. Uguz is partially supported by TUBITAK 110T713.

References

[1] Bovik A. Handbook of image and video processing. New York: Academic; 2000.
[2] Srinivasan KS, Ebenezer D. A new fast and efficient decision-based algorithm for removal of high-density impulse noises. IEEE Signal Process Lett 2007;14:189–92.
[3] Nodes TA, Gallagher NC. The output distribution of median type filters. IEEE Trans Commun 1984;32(5):532–41.
[4] Luo W. Efficient removal of impulse noise from digital images. IEEE Trans Consum Electron 2006;52(2).
[5] Hwang H, Haddad RA. Adaptive median filters: new algorithms and results. IEEE Trans Image Process 1995;4(4):499–502.
[6] Nair MS, Raju G. Additive noise removal using a novel fuzzy-based filter. Comput Electr Eng 2011;37:644–55.
[7] Toh KKV, Isa NAM. Noise adaptive fuzzy switching median filter for salt-and-pepper noise reduction. IEEE Signal Process Lett 2010;17(3):281–4. http://dx.doi.org/10.1109/LSP.2009.2038769.
[8] Eng HL, Ma KK. Noise adaptive soft-switching median filter. IEEE Trans Image Process 2001;10(2):242–51.
[9] Akkoul S, Ledee R, Leconge R, Harba R. A new adaptive switching median filter. IEEE Signal Process Lett 2010;17(6):587–90.
[10] Pok G, Liu JC, Nair AS. Selective removal of impulse noise based on homogeneity level information. IEEE Trans Image Process 2003;12(1):85–92.
[11] Nair MS, Mol PMA. Direction based adaptive weighted switching median filter for removing high density impulse noise. Comput Electr Eng 2013;39(2):663–89.
[12] Wolfram S. Universality and complexity in cellular automata. Physica 1984:1–35.
[13] Selvapeter PJ, Hordijk H. Cellular automata for image noise filtering. In: World congress on Nature Biologically Inspired Computing (NaBIC); 2009. p. 193–197.
[14] Popovici A, Popovici D. Cellular automata in image processing. In: Gilliam DS, Rosenthal J, editors. Proceedings of the 15th international symposium on the mathematical theory of networks and systems; 2002.
[15] Rosin PL. Training cellular automata for image processing. IEEE Trans Image Process 2006;15(7):2076–87.
[16] Rosin PL. Image processing using 3-state cellular automata. Comput Vis Image Understand 2010;114:790–802.
[17] Songtao L, Chen H, Yang S. An effective filtering algorithm for image salt-pepper noises based on cellular automata. In: Congress on image and signal processing, CISP '08, vol. 3; 2008. p. 294–97.
[18] Sadeghi S, Rezvanian A, Kamrani E. An efficient method for impulse noise reduction from images using fuzzy cellular automata. Int J Electron Commun (AE) 2012:772–9.
[19] Toh KKV, Ibrahim H, Mahyuddin MN. Salt-and-pepper noise detection and reduction using fuzzy switching median filter. IEEE Trans Consum Electr 2008;54:1956–61.
[20] Siap I, Akın H, Uguz S. Structure and reversibility of 2D hexagonal cellular automata. Comput Math Appl 2011;62:4161–9.
[21] Wang Z, Bovik AC, Sheikh HR, Simoncelli EP. Image quality assessment: from error visibility to structural similarity. IEEE Trans Image Process 2004;13(4):600–12.
[22] Hashimoto Y, Kajikawa Y, Nomura Y. Directional difference based switching median filters. Electron Commun Jpn 2002;85:22–32.
[23] Zhang S, Karim MA. A new impulse detector for switching median filters. IEEE Signal Process Lett 2002;9(11):360–3.
[24] Wang Z, Zhang D. Progressive switching median filter for the removal of impulse noise from highly corrupted images. IEEE Trans Circ Syst II, Analog Digit Signal Process 1999;46(1):78–80.
[25] Lu CT, Chou TC. Denoising of salt and pepper noise corrupted image using modified directional weighted median filter. Patt Recogn Lett 2012;33(10):1287–95.
[26] Wang SS, Wu CH. New impulse detection and filtering method for removal of wide range impulse noises. Patt Recogn 2009;42:2194–202.

[27] Chan RH, Ho CW, Nikolova M. Salt and pepper noise removal by median type noise detectors and detail preserving regularization. IEEE Trans Image Process 2005;14(10):1479–85.

Ugur Sahin received his Ph.D. at Istanbul Technical University (ITU), Turkey in 2010. His current position is Post-Doctorate Fellow at Multi Agent Bio Robotic Laboratory at Rochester Institute of Technology. His current research interests includes Fuzzy Logic, Cellular Automata, Bayesian Networks and Image Processing applications.

Selman Uguz received his Ph.D. at Istanbul Technical University (ITU), Turkey in 2009. Since March 2009, he has been working at Harran University, Department of Mathematics, Sanliurfa, Turkey. His current research interests includes n-dimensional Cellular Automata and Image Processing applications.

Ferat Sahin received his B.Sc. in Electronics and Communications Engineering from Istanbul Technical University, Turkey, in 1992 and M.Sc. and Ph.D. degrees from Virginia Polytechnic Institute and State University in 1997 and 2000, respectively. In September 2000, he joined Rochester Institute of Technology, where he is an Associate Professor. He is also the director of Multi Agent Bio-Robotics Laboratory at RIT. His current research interests are System of Systems, Swarm Intelligence, Robotics, MEMS Materials Modeling, MEMS-based Microrobots, Micro Actuators, Distributed Computing, Decision Theory, Pattern Recognition, Distributed Multi-agent Systems, and Structural Bayesian Network Learning. He has over 70 conferences and journals in these areas.